

NOVA IMS
BSC IN DATA SCIENCE
BIG DATA ANALYSIS

Big Data Analysis in Spark

Final Project

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1 Introduction

Big data has revolutionized how businesses use information, presenting tools to deal with the 4 V's of Big Data: Volume, Variety, Velocity, and Veracity. To address these, strong frameworks that enable efficient data processing, analysis, and actionable insights are needed. With its powerful and scalable answer to big data problems, Apache Spark becomes an essential tool for enterprises. This report describes how to use Spark's capabilities to tackle a variety of problems using different datasets, while also delivering highly relevant insights.

2 Analysis Overview

The following tasks were carried out using these three datasets:

- **Student Performance Dataset:** Perform large-scale exploration and cleaning; student segmentation, so schools can develop different educational processes for them; network analysis.
- **Musical Data Dataset:** Perform large-scale and extensive data exploration and cleaning; machine learning pipelines to predict song popularity; cross-validation hyperparameter tuning.
- **Loan Approval Dataset:** Perform large-scale exploration and cleaning; machine learning pipelines to predict loan approvals; analyze real-time data with streaming.

3 Methodology

3.1 Pre-Processing and Exploratory Data Analysis:

The initial pre-processing and data cleaning involved checking for duplicates, imputing missing values, and correcting data types using a user-defined function. Possible inconsistencies were checked, and descriptive statistics analyzed. A thorough data exploration was performed, supported by visualizations. Spark's **RDDs** facilitated scalable distributed operations across partitions, enhancing efficiency for large datasets, and **DataFrames** were used for operations such as `groupBy`, `agg`, `filter` and `select`, enabling efficient querying and processing. **Spark SQL** was also helpful for querying. To optimize performance with large datasets, operations like `orderBy` and `collect` were avoided whenever possible.

3.1.1 Use Case: Student Performance

- **Tasks:** Data cleaning and exploration were carried out, including an analysis of variable distributions, correlations and other techniques to uncover insights.
- **Insights:** Analysis of all variables revealed that most students are from Gabriel Pereira School, predominantly urban, with families larger than three members. The majority have parents living together, receive family support, and aim for higher education. Mothers are the typically primary guardians, parental employment is mainly in services or so called other professions, and school choice is influenced by reputation and proximity. Additionally, there were 2,255 student absences, with around 83 percent having 10 absences or more. Regarding school comparison, we found that students from Gabriel Pereira School had significantly higher alcohol consumption compared to Mousinho da Silveira School. Lastly, strong correlations were found between the 1st, 2nd, and final grades, moderate correlation between parents' education levels, and a positive correlation between studytime and grades. More failures and absences were linked to lower grades, while age and health had minimal impact.

3.1.2 Use Case: Musical Data

- **Tasks:** Data cleaning was performed, inconsistencies were treated, and data exploration, covering artist, song, stream, platform, time, and correlation analysis was employed.

- **Insights:** Identified the most and least present artists based on song count across platforms. Top artists such as Taylor Swift, The Weeknd, and Bad Bunny were found at the front, while Yoasobi and Beach House were among the least present. The top artists typically had songs with a moderate BPM and low acousticness, while the less-present artists showed higher BPM, lower danceability, and greater instrumentalness. We verified that top songs varied across the charts analyzed (in overall playlists, charts, and streams), highlighting varying criteria for them, and that songs with 1-2 artists tend to have higher streams. Regarding collaborations, 21 Savage and Drake were the most frequent artist pair, with 20 songs together. Release trends showed 2022 as the most common year, January as the top month, and the 7th as the most frequent day, with no specific explanation found for that event. Spotify led in playlist presence, followed by Deezer and Apple. Finally, no strong correlations between streams and musical features were found.

3.1.3 Use Case: Loan Approval

- **Tasks:** Data cleaning was performed, inconsistencies were checked and extreme values treated, and data exploration was employed, covering variable distributions and correlation analysis.
- **Insights:** We found discrepancies in loan applications, such as suspiciously high ages, thus we concentrated on ages less than 95 for reliability. We also discovered people with salaries of more than 1 million seeking for minor loans, which seemed inconsistent, as well as candidates with exceptionally high years of work experience. Regarding loan approval, we discovered that rejections are more common than approvals. Individuals with only a high school, bachelor's, or associate degree are refused more frequently than those with a higher education, and people paying mortgages or rent are more likely to be denied than homeowners. In terms of loan intent, medical requirements are the most typical cause for approval, followed by debt consolidation. Regarding variable correlations we found that larger loans align with higher income percentages, and higher credit scores lead to lower interest rates.

3.2 Comprehensive Machine Learning Workflows:

We demonstrated Spark's ability to streamline machine learning processes, and of using integrated pipelines. Tasks like feature engineering, model training, validation, and performance evaluation were performed, ensuring scalability and consistency.

3.2.1 Use Case: Student Performance

- **Tasks:** Used **K-Means Clustering** to divide students into clusters based on their attributes, using the elbow method to determine the optimal number of clusters. Further investigation was done with **Graphframes** and network visualizations.
- **Results:** Found distinct student segments based on academic performance and behavior. One segment is characterized by lower levels of parental education, slightly above average study time, and varied home to school travel times. There is a high presence of romantic relationships, and lower parental job influence. The other group, has higher levels of parental education, particularly from the mother's side. These students have lower study and travel times, and they display stronger motivations for studying. We investigated student social behaviors, family, and academic background networks. We discovered moderately high social connectivity among students, with stronger connections between those with similar family backgrounds or academic performances. These high levels are most likely the result of all students attending the same two schools, who share similar features within that setting.

3.2.2 Use Case: Musical Data

- **Tasks:** Apply several classification models with the use of **pipelines**, such as **Logistic Regression**, **Random Forest**, and **Gradient Boosting Trees**, with the goal of predicting a song's popularity based on its musical attributes. To fine-tune hyper parameters **cross-validation** was applied to these models.
- **Results:** The best model was Random Forest, with an AUC score of around 60 percent, indicating that predicting a song's popularity based solely on its features is unreliable due to uncontrollable factors. However, as mentioned, previous exploratory analysis revealed key components commonly found in popular songs, which could help artists in creating their songs.

3.2.3 Use Case: Loan Approval

- **Tasks:** Apply several classification models with the use of **pipelines**, such as **Logistic Regression**, **Random Forest**, **Gradient Boosting Trees**, and **Support Vector Machines**, with the goal of predicting whether a loan application is approved or not based on its candidate characteristics. Moreover, **Spark Streaming** was employed to demonstrate the value of real-time analytics in financial decision-making, by handling simulated real-time data for loan approval and generating queries from it.
- **Results:** The best model was Gradient-Boosted Tree Classifier, with an AUC score of approximately 95 percent. The model's excellent performance indicates that its features, such as income, credit history, and other applicant attributes, have a considerable impact on loan approvals. As a result, this model may be trusted to produce accurate forecasts, giving vital insights for financial institutions in their decision-making process. Finally, we found querying real-time data highly beneficial and recommend this strategy to help prevent potential fraud.

4 Conclusion

This report demonstrates the versatility and power of Apache Spark in dealing with Big Data to gain insights. By utilizing Spark's diverse functionalities, from RDDs to machine learning pipelines and streaming, we showcase its potential to deliver scalable, efficient, and actionable data solutions. Implementing Spark in organizational workflows empowers management with insights drawn from vast and varied datasets, unlocking new opportunities for innovation and strategic growth.

5 References

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